

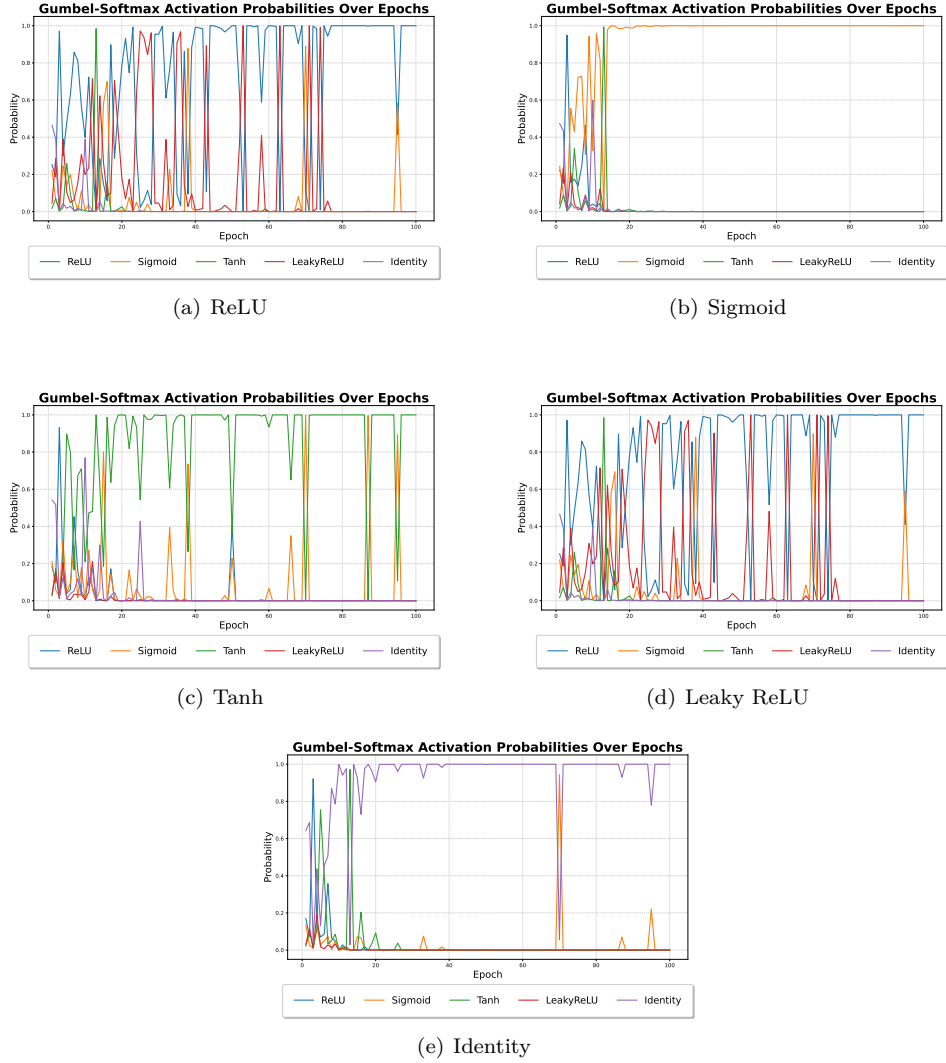


Figure 1: Gumbel-Softmax activation selection probabilities over training epochs for various ground-truth activation functions. The model consistently converges to the appropriate non-linearity across tasks.

trick and a novel gradient-norm-based regularizer, our method effectively learns which non-linearity is most appropriate for the task, without requiring architecture-specific tuning or manual hyperparameter sweeps. In controlled experiments where the ground-truth function is known, **Flex-Act** consistently recovered the optimal activation while maintaining interpretability and modularity.

Our results highlight a compelling possibility: the same network architecture can be reused across a diverse range of functional regimes—each requiring distinct activation dynamics—without retraining or redesign. This opens the door to more robust, task-adaptive neural networks where non-linearity is no longer a fixed hyperparameter but a learnable component of the model architecture.

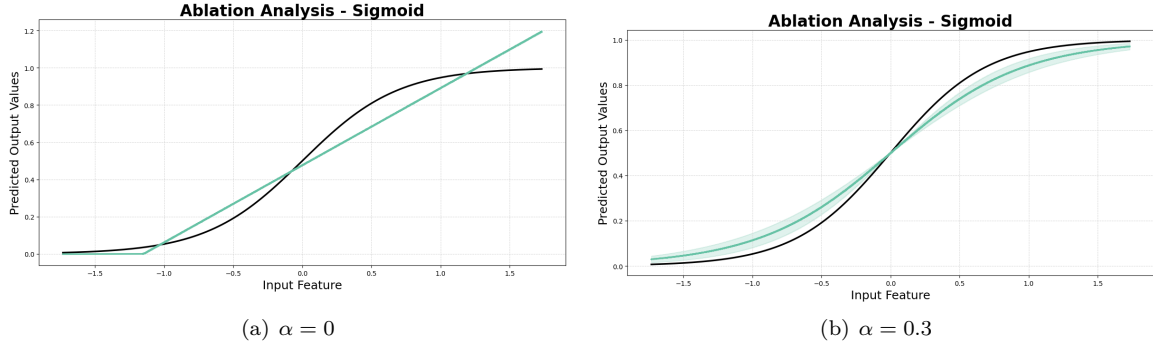


Figure 2: Effect of KL-based regularization on activation selection. Without regularization, the model biases toward ReLU. With regularization, convergence improves significantly.

Broader Impact Statement

The proposed framework for discrete activation routing has implications for the design and deployment of more adaptive and efficient neural networks. By enabling a single architecture to automatically adjust its activation functions to suit diverse learning tasks, **Flex-Act** reduces the need for manual tuning and hyperparameter searches in this context. This can lower both the computational cost and the barrier to entry for training high-performing models across domains.

We do not foresee direct negative societal impacts of this work. However, as with any general-purpose machine learning method, downstream misuse is always a possibility. To mitigate risks, we encourage the community to pair technical progress with domain-specific oversight when deploying adaptive models in sensitive applications. Furthermore, our method preserves interpretability, which supports transparency and post hoc auditing in high-stakes settings.

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